

## ELEMENTS AND COMPOUNDS

**Element:** a substance that cannot be broken down into simpler substances and contains all the same type of atoms.

- There are 92 naturally occurring elements and 24 manufactured elements (2005 data).
- Each element has its own unique atoms - no two elements have the same atoms.
- Elements are given symbols to represent them.
- The symbols for elements always start with a capital letter, and the second letter (if used) is always lower case.

e.g. Al - aluminium C - carbon

- The symbol is based on the English name, though some are based on Latin, Arabic and German names.

| NON-ENGLISH |    |
|-------------|----|
| sodium      | Na |
| potassium   | K  |
| iron        | Fe |
| copper      | Cu |
| silver      | Ag |
| tungsten    | W  |
| gold        | Au |
| mercury     | Hg |
| lead        | Pb |
| tin         | Sn |

|           |    |
|-----------|----|
| cobalt    | Co |
| manganese | Mn |
| bromine   | Br |
| iodine    | I  |
| zinc      | Zn |
| nickel    | Ni |
| barium    | Ba |

### ENGLISH

|            |    |
|------------|----|
| hydrogen   | H  |
| helium     | He |
| lithium    | Li |
| beryllium  | Be |
| boron      | B  |
| carbon     | C  |
| nitrogen   | N  |
| oxygen     | O  |
| fluorine   | F  |
| neon       | Ne |
| magnesium  | Mg |
| aluminium  | Al |
| silicon    | Si |
| phosphorus | P  |
| sulfur     | S  |
| chlorine   | Cl |
| argon      | Ar |
| calcium    | Ca |

These are most of the first 20 elements of the Periodic Table.

**Compound:** two or more elements chemically combined.

- Most substances are compounds.
- Some common substances are given below.

|                   |                                |                 |                   |
|-------------------|--------------------------------|-----------------|-------------------|
| hydrochloric acid | HCl                            | carbon dioxide  | CO <sub>2</sub>   |
| sulphuric acid    | H <sub>2</sub> SO <sub>4</sub> | oxygen gas      | O <sub>2</sub>    |
| nitric acid       | HNO <sub>3</sub>               | copper sulphate | CuSO <sub>4</sub> |

### Consider H<sub>2</sub>SO<sub>4</sub>

- The 3 capital letters represent 3 elements - hydrogen (H), sulphur (S) and oxygen(O).
- The numbers represent how many atoms of each element are present.

e.g. There are 2 x H atoms, 1 x S atom, and 4 x O atoms.

### Examples

| NAME               | FORMULA                                         | DESCRIPTION                                         |
|--------------------|-------------------------------------------------|-----------------------------------------------------|
| calcium carbonate  | CaCO <sub>3</sub>                               | 1 x Ca atom, 1 x C atom, 3 x O atom                 |
| acetic acid        | CH <sub>3</sub> COOH                            | 2 x C atoms, 2 x O atoms, 4 x H atoms               |
| zinc hydroxide     | Zn(OH) <sub>2</sub>                             | 1 x Zn atom, 2 x O atoms, 2 x H atoms               |
| ammonium phosphate | (NH <sub>4</sub> ) <sub>3</sub> PO <sub>4</sub> | 3 x N atoms, 12 x H atoms, 1 x P atom, 4 x O atoms. |

The 3 multiplies everything in the bracket.

i.e. 3 x 4 x H = 12 x H atoms  
3 x N atoms